No Spoiler Please! A TinyBERT Model to Detect Spoiler Reviews on IMDb

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Abstract

This report contains the explanation of the work done for the final project of the Natural Language Processing course. The professor in charge of the course is Professor Alfio Ferrara, University of Milan. The project focused on the application of knowledge distillation, a technique used to compress large language models into smaller and more efficient versions. This is achived by training a smaller model (the student) to replicate the behavior of a larger model (the teacher). In particular, this project had the aim to implement a spoiler detector for IMDb's movie reviews: a BERT-base model has been used as the teacher model, while a TinyBERT_General_4L_312D has been used as the student model. The experiments shows that WAITING FOR RESULTS

1 Introduction

Very often, before even seeing a film, we decide to read some reviews because we're curious to know the opinions of those who have already seen it. In such a situation, it's not uncommon to encounter reviews that, whether intended to spoil something or not, include spoilers (e.g., "The movie is great, but character X dies/does this or that"). Clearly, we want to avoid this: luckily, platforms like IMDb warn us in advance about such reviews.

IMDb is by far the most popular global source for movies and TV shows, and it also includes user-submitted reviews with spoiler warnings for each one. The problem is that users label reviews as spoilers themselves, but they can lie and insert spoilers on a no spoiler marked review. Other users can report a review as spoiler, and the IMDb moderation team will label it as such, but of course it's not immediate. What if there were a browser extension (running locally on users' PCs) that retrieves the text of a review and tells us "Spoiler alert!" before we can even begin reading it? With the recent evolution of LLMs, this task can be accomplished by fine-tuning a sufficiently large pre-trained model, such as BERT.

1.1 Related Work

2 Research question and methodology

2.1 Proposed approach

3 Experimental Results

3.1 Data and Preprocessing

3.2 BERT Fine Tuning for Spoiler Detection

3.2.1 Results of BERT Fine Tuning

3.3 Task-Specific Distillation

3.3.1 Phase 1: Intermediate Layers Distillation

3.3.2 Phase 2: Prediction Layer Prediction

4 Results of distillation

4.1 Performance Drop

4.2 Computational Savings

4.3 Interpretability

5 Conclusions

References